Practical 3b: Simple Web API returning Student data in JSON

AIM : To create a simple ASP.NET Core Web API that returns Student data in JSON format.

Models/Student.cs

```
namespace StudentAPI.Models
{
   public class Student
   {
      public int Id { get; set; }
      public string Name { get; set; }
      public string Course { get; set; }
      public int Age { get; set; }
      public double Percentage { get; set; }
   }
}
```

Controllers/StudentsController.cs

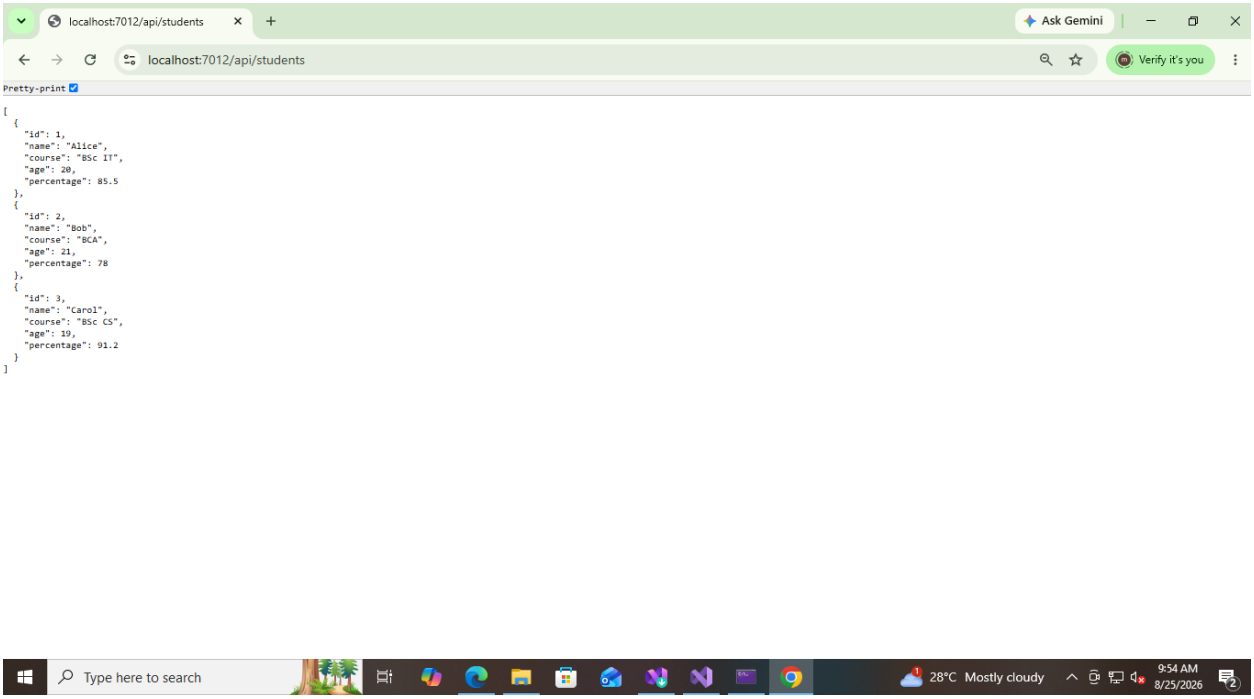
```
using Microsoft.AspNetCore.Mvc;
using StudentAPI.Models;

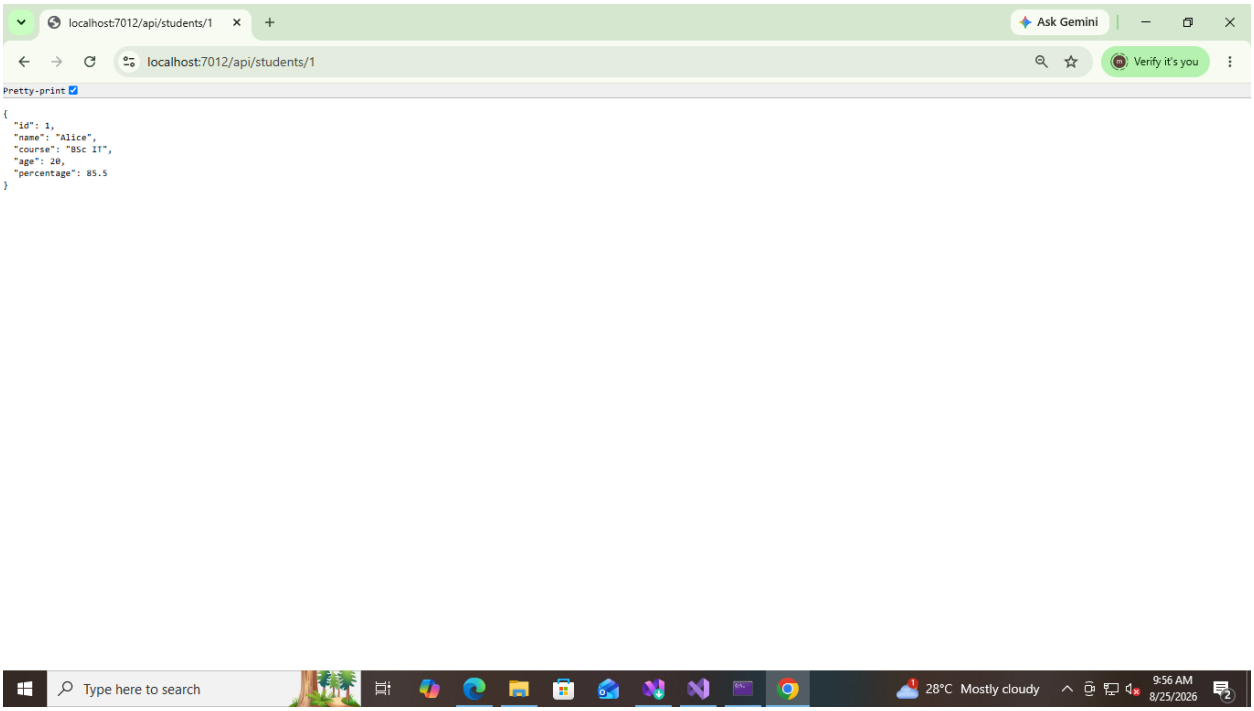
namespace StudentAPI.Controllers
{
   [ApiController]
   [Route("api/[controller]")]
   public class StudentsController : ControllerBase
   {
      private static List<Student> students = new List<Student>
      {
         new Student { Id = 1, Name = "Alice", Course = "BSc IT", Age = 20, Percentage = 85.5
},
         new Student { Id = 2, Name = "Bob", Course = "BCA", Age = 21, Percentage = 78.0 },
         new Student { Id = 3, Name = "Carol", Course = "BSc CS", Age = 19, Percentage = 91.2
}
      };
```

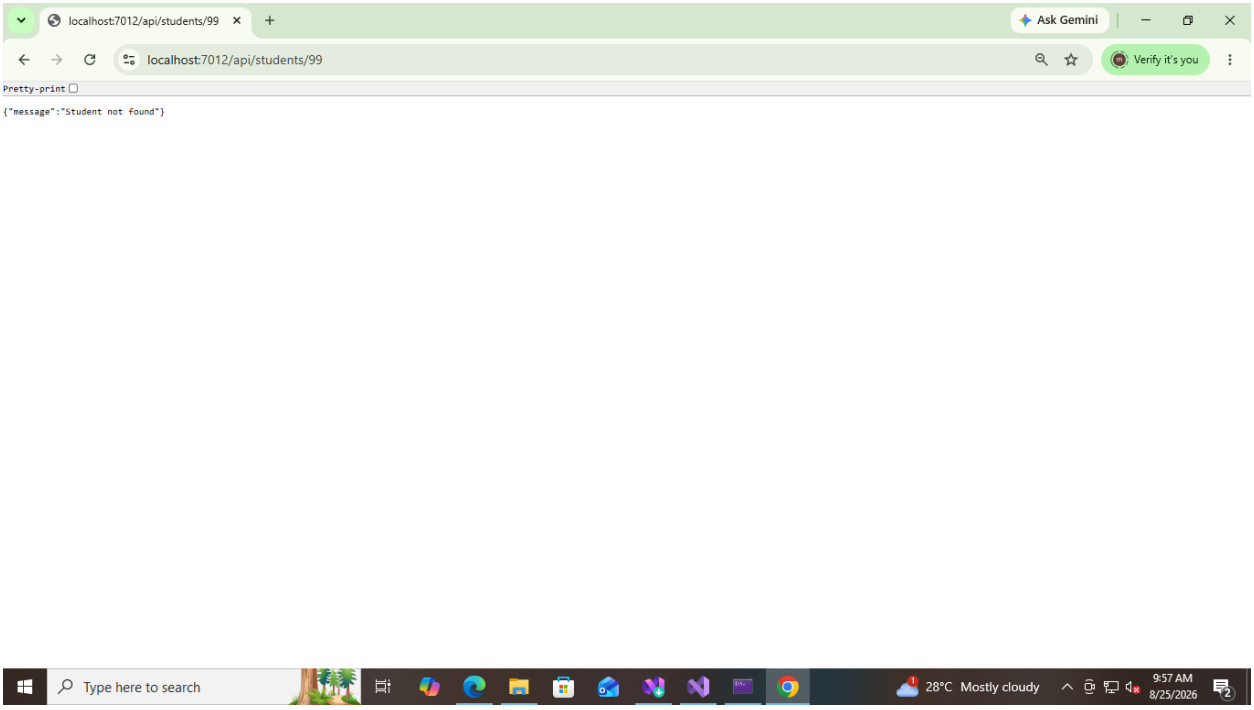
```
// GET: api/students
        [HttpGet]
        public ActionResult<IEnumerable<Student>> GetAll()
            => Ok(students);

        // GET: api/students/1
        [HttpGet("{id}")]
        public ActionResult<Student> GetById(int id)
        {
            var s = students.FirstOrDefault(x => x.Id == id);
            if (s == null) return NotFound(new { message = "Student not found" });
            return Ok(s);
        }
    }
}
```

Output:




localhost:7012/api/students/1
Ask Gemini
localhost:7012/api/students/1
Verify it's you
Pretty-print
{
"id": 1,
"name": "Alice",
"course": "BSc IT",
"age": 20,
"percentage": 85.5
}
Type here to search
28°C Mostly cloudy
9:56 AM
8/25/2026


localhost:7012/api/students/99
Ask Gemini
localhost:7012/api/students/99
Verify it's you
Pretty-print
{"message":"Student not found"}
Type here to search
28°C Mostly cloudy
9:57 AM
8/25/2026